A ZX SPECTRUM HANDHELD CONCEPT

# A whole era. A tiny form.

Meet Speccy micro. A compact, USB-powered design for a very familiar kind of play.

**24 × 38 × 12 MM**

Modeled envelope, including button caps.

EXPLORE THE CONCEPT →




Concept visualization · illustrative screen

THE DIMENSIONS ARE THE DESIGN

# Small is the point.

A complete modeled enclosure in a footprint of 24 × 38 mm. The 12 mm depth includes the six button caps; the USB cable is separate.

| 24 | 38 | 12 |
|---|---|---|
| WIDTH / MM | LENGTH / MM | DEPTH / MM |

Dimensions are a packaging target; physical fit is open.

38 MM




ENLARGED CONCEPT VIEW

THE SIZE OF THE IDEA

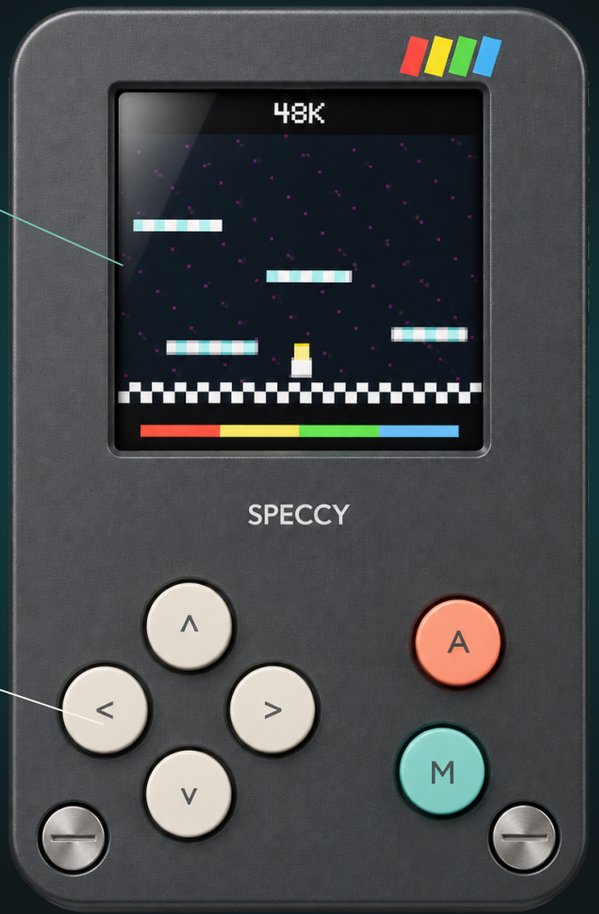



Print at 100% for actual scale.

DESIGNED EXPERIENCE

# A familiar way to play.

The selected workload is original Dizzy on Spectrum 48K, using Kempston inputs.




## Four directions

Up, Down, Left and Right.

## Fire

The game’s action button.

## Menu

Pause, Resume and Restart.

Concept visualization · physical controls remain untested.

THE PICTURE, RECONSIDERED

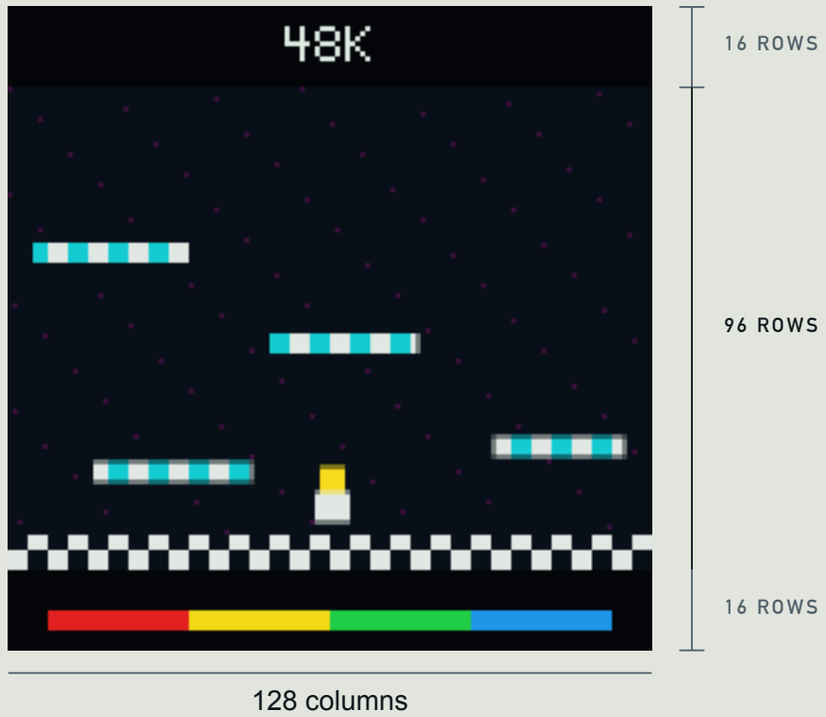
# 128 pixels. Still Spectrum.

The 256 × 192 game picture is reduced by averaging each 2 × 2 pixel block. The result is 128 × 96, centered on a 128 × 128 LCD.

**2:1** REDUCTION

Compactness has a visible tradeoff: fine detail and text lose clarity. Readability on the actual display remains a prototype test.

128 × 128 / DISPLAY GRID



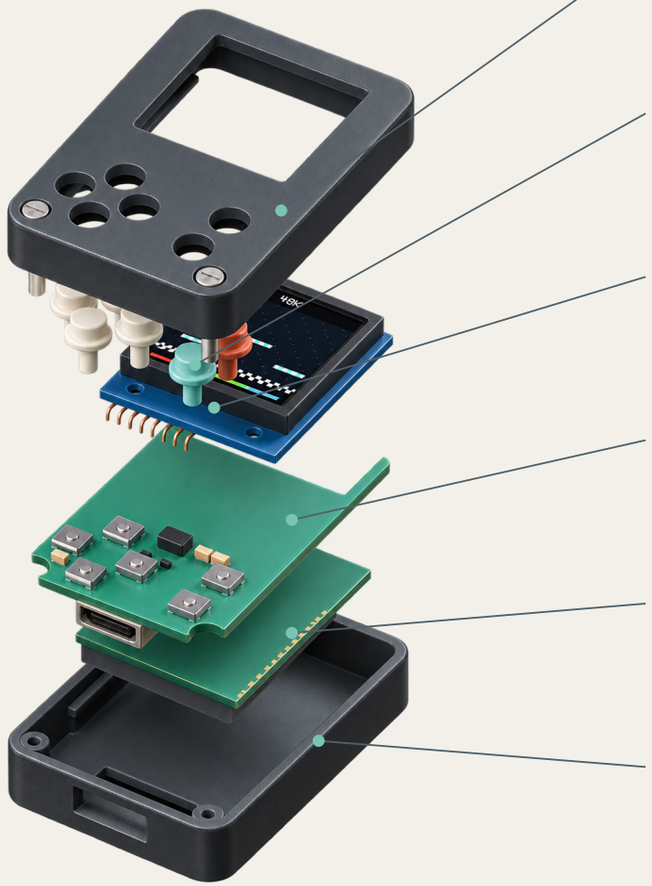

CAD artwork shown; firmware uses border-colored rows.

Illustration, not a game or runtime capture.

THE MECHANICAL STACK

# Every layer has a reason.

Six layers in the presentation.
One compact assembled envelope.



06 **Front shell**
Screen window and control openings.

05 **Six button plungers**
Four directions, Fire and Menu.

04 **0.85-inch LCD**
Connector replaced by eight wires.

03 **Custom carrier**
Switches, power and external UART.

02 **ESP32-WROOM-32E-N4**
The selected module below the carrier.

01 **Rear shell**
USB opening and programming access.

THE STACK STILL NEEDS SAMPLES

LCD depth, switch travel, wires and socket clearance need measured parts and assembly trials.

Exploded concept visualization · layer spacing is illustrative.

THE SELECTED HARDWARE

# Small outside. Specific inside.

| | |
|---|---|
| PROCESSOR | ESP32-WROOM-32E-N4 |
| DISPLAY | Waveshare 0.85″ / GC9107 / SPI |
| POWER | Micro-B 5 V / 3.3 V LDO |
| CONTROLS | Four directions / Fire / Menu |
| CONTENT | 48K ROM + selected SNA in flash |
| PROGRAMMING | External UART / six underside pads |

USB-powered · battery-free · radios disabled



UNDERSIDE / SERVICE INTERFACE

The fixture uses manual BOOT / RESET and gated UART. Its 3V3 pogo pad senses the rail; it does not supply the device.

Concept view · fixture operation awaits bench tests

RECORDED REPOSITORY BASELINE

# A concept with real evidence.

The digital and software work is tangible.
A working physical device is still the next proof.

## Digital geometry

47 native shapes checked; eight closed STL meshes.
The modeled envelope has been verified.

## Firmware

Three native test suites pass; two ESP32 modes build.
Original Dizzy starts, walks and jumps on the host.

## Electrical study

Carrier and fixture pass electrical rule checks.
The placement board is unrouted; DRC remains open.

Recorded evidence, not new hardware test results.

OPEN / PHYSICAL VALIDATION

## Next proof: the real device.

- **Fit & assembly**
  Switch travel, USB clearance, LCD and wires.
- **Power & heat**
  Current, source limits and enclosure temperature.
- **Play & readability**
  Display, controls and real-time performance.
- **Programming & recovery**
  Probe access, boot, flash and interrupted power.

A SMALL DESIGN. A CLEAR NEXT STEP.

# Make the next millimetre count.

Explore the source. Measure the selected parts.
Turn the digital study into evidence on the bench.




REVIEW THE EVIDENCE →

EXPLORE THE SYSTEM →

SOURCE NOTES / IN THE REPOSITORY

**Product & verification**

docs/product-brief.md · docs/verification.md

**CAD, firmware & electronics**

hardware/cad/ · firmware/ · hardware/electronics/

Concept prospectus · physical acceptance, price, release date and distribution rights remain open.